

Optimizing Diabetic Retinopathy Detection: A Progressive Methodology Using GeM Pooling and Ordinal Regression

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Abstract

The automated grading of Diabetic Retinopathy (DR) requires sensitivity to subtle anatomical markers and robustness against severe class imbalances. This report details a multi-phase optimization of an EfficientNet-B3 architecture on the APTOS 2019 dataset. We demonstrate that combining aggressive contrast-enhancing V2 augmentation with targeted minority oversampling and Focal Loss significantly stabilizes training. By replacing standard global average pooling with Generalized Mean (GeM) pooling ($p = 3$), the network effectively localizes high-intensity disease markers. Finally, we implement Conditional Ordinal Regression for Neural Networks (CORN) to align the optimization objective with the progressive severity of the disease. Our un-frozen phase three GeM model achieved a peak Candidate Quadratic Weighted Kappa (QWK) of 0.8841, while our fully un-frozen CORN model achieved a highly competitive QWK of 0.8831.

1 Introduction

Diabetic Retinopathy is graded on a progressive, ordinal scale ranging from 0 (No DR) to 4 (Proliferative DR). Standard categorical classification penalizes all incorrect predictions equally, failing to capture the clinical reality that misclassifying a Grade 4 as a Grade 0 is a fundamentally more severe error than misclassifying it as a Grade 3. Our methodology addresses this through four evolutionary phases: establishing a frozen EfficientNet-B3 base-

line, engineering a robust augmentation and oversampling pipeline, fine-tuning with GeM pooling, and transitioning to an ordinal regression framework.

2 Data Pipeline and Preprocessing

The APTOS 2019 dataset features a heavy class imbalance, heavily skewed toward Grade 0 (No DR).

2.1 Oversampling Strategy

To prevent the model from collapsing into majority-class predictions, we implemented a deterministic array-tiling oversampling strategy. By analyzing the class distribution, we applied explicit multipliers to the minority classes: Grade 1 (Mild) was oversampled by a factor of 2, Grade 3 (Severe) by a factor of 5, and Grade 4 (Proliferative) by a factor of 3, while Grade 2 remained un-multiplied. This rebalanced the batches for the data loader.

2.2 V1 vs. V2 Augmentation

Initial preprocessing involved a standard V1 augmentation pipeline (horizontal flips, 90-degree rotations, and minor brightness adjustments). We also experimented with Graham’s color normalization; however, because the APTOS images possess inherently good lighting, Graham’s method deteriorated performance. The base model with V1 augmentation yielded an F1 score of 0.5730 and a validation loss of 0.3000 over 10 epochs.

To force the network to become invariant to lighting and camera differences, we developed an aggressive V2 pipeline using Albumentations. This included affine transformations (scaling 0.8-1.2, translating ± 0.2 , full 360-degree rotations, and shear), HSV shifts (Hue shift limit 10, Saturation shift 20), and dynamic blurring/sharpening. Crucially, the V2 pipeline strips natural retinal hues to drastically increase the contrast of microaneurysms and hemorrhages. All images were subsequently normalized using ImageNet means (0.485, 0.456, 0.406) and standard deviations (0.229, 0.224, 0.225).

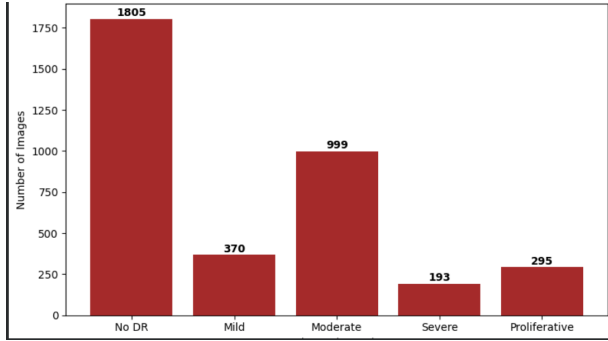


Figure 1: Class distribution of the APTOS 2019 dataset before augmentation.

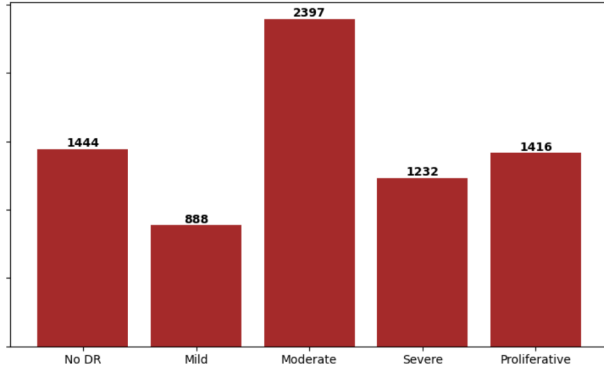


Figure 2: Class distribution after applying targeted array-tiling oversampling.

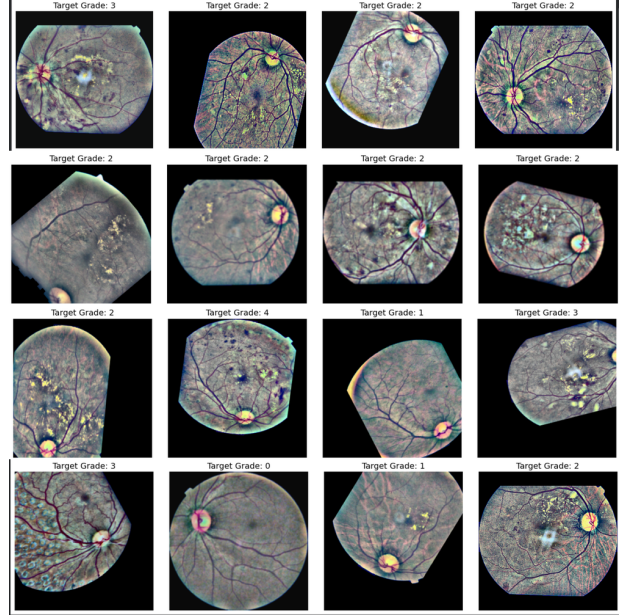


Figure 3: Sample fundus images processed through the aggressive V2 Augmentation pipeline.

3 Architectural Modifications

3.1 Focal Loss

Before transitioning to ordinal regression, we replaced standard Cross-Entropy with Focal Loss ($\gamma = 2.0$) to dynamically scale the loss based on prediction confidence. By reducing the weight of easy, majority-class examples, the network focused its gradients on hard-to-classify edge cases.

3.2 Generalized Mean (GeM) Pooling

Standard Global Average Pooling (GAP) uniformly averages all spatial features, which can dilute the activation signal of tiny, localized lesions in a large 384x384 image. We replaced GAP with a Generalized Mean (GeM) pooling layer, defined mathematically as:

$$y = \left(\frac{1}{N} \sum_{i=1}^N x_i^p \right)^{\frac{1}{p}}$$

We initialized the pooling parameter at $p = 3$. Because $p > 1$, the operation behaves closer to max-pooling, forcing the network to highlight the most intense pathological features rather than averaging the entire retina. The frozen GeM model with V2 augmentation improved baseline metrics significantly, reaching a QWK of 0.8495 and an F1 of 0.6125 over 40 epochs.

4 Phase 3: Fine-Tuning Strategy

To adapt the EfficientNet-B3 backbone specifically to diabetic markers, we executed a two-stage unfreezing protocol. We unfroze the top three layers of the network, activating approximately 8.5 million parameters (nearly 80% of the model’s total capacity).

To stabilize this massive influx of trainable parameters and prevent overfitting, we utilized the Adam optimizer with a conservative learning rate of $1e-3$ and a weight decay of $1e-4$. We incorporated a ‘ReduceLROnPlateau’ scheduler (factor=0.5, patience=2) and an ‘EarlyStopping’ monitor tracking maximum validation performance (patience=5, min_delta=0.001). Furthermore, the input image resolution was scaled up to 384x384, and the data split was adjusted to a strict 80/20 ratio.

The resulting unfrozen Candidate GeM model (at resolution 300) achieved our highest overall QWK of 0.8841 and an F1 of 0.6919. The Champion GeM model (at resolution 384, using $5e-4$ weight decay for better generalization) stabilized at a QWK of 0.8774 and an F1 of 0.6488.

5 Phase 4: Ordinal Regression (CORN)

5.1 Intuition and Architecture

The inherent flaw in our Phase 3 models was the use of categorical loss, which treats DR grades as independent, unrelated classes. To encode the progressive nature of the disease, we implemented Conditional Ordinal Regression for Neural Networks (CORN).

Instead of outputting 5 independent class probabilities, the CORN classification head outputs $K - 1$ (in this case,

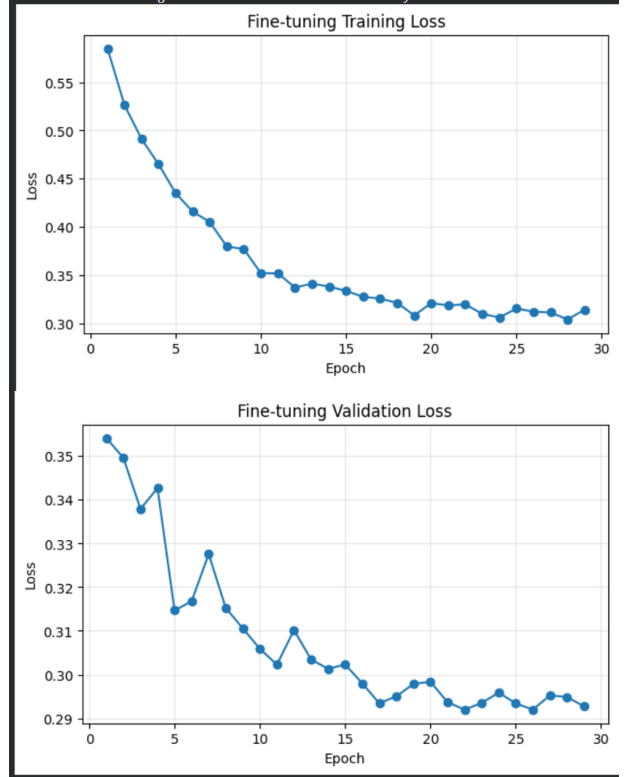


Figure 4: Training and validation loss curves exhibiting stable convergence for the unfrozen GeM model.

4) conditional binary probabilities. The network asks sequential questions: “Is the disease worse than Grade 0?”, “Is it worse than Grade 1?”, and so forth. This forces the network to learn a unified threshold for disease severity.

5.2 CORN Results

We trained the CORN models utilizing the same robust optimizer and scheduling parameters established in Phase 3. The initial frozen Corn V1 model (using V1 augmentation) achieved a QWK of 0.8430 and an F1 of 0.5988. Upgrading to V2 augmentation with the GeM layer (Corn V2, frozen) yielded a QWK of 0.8532 and an F1 of 0.5735.

Our final, fully unfrozen CORN model (384 resolution, 80% training split, $5e-4$ weight decay) successfully merged the geometric stability of the V2 pipeline with the

mathematical rigor of ordinal loss. Over 40 epochs, it achieved a final QWK of 0.8831 and an F1 of 0.6338, accompanied by a highly convergent training loss of 0.2255 and validation loss of 0.2863.

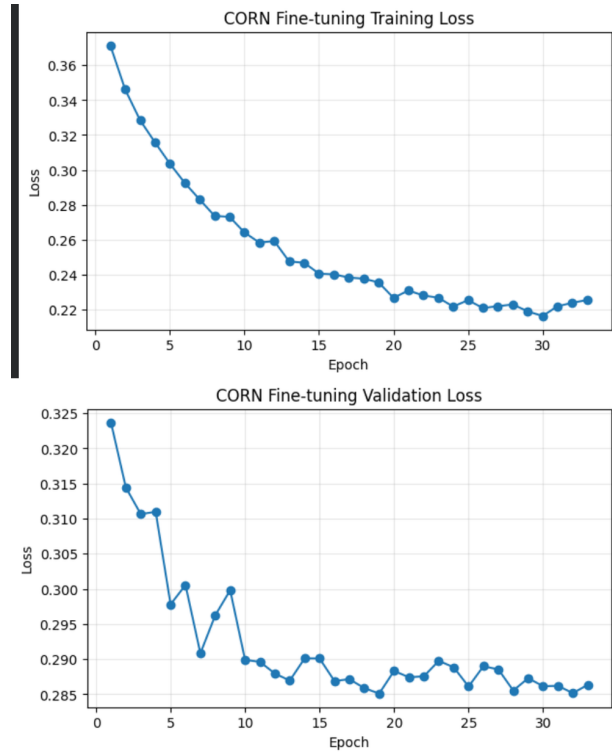


Figure 5: Training and validation diagnostics for the Final CORN Model.

6 Interpretability

To verify that the network localized valid pathological features rather than exploiting background artifacts, we utilized Grad-CAM (Gradient-weighted Class Activation Mapping). The resulting heatmaps confirmed the unfrozen GeM layer successfully directed the network’s attention toward critical microaneurysms and exudates.

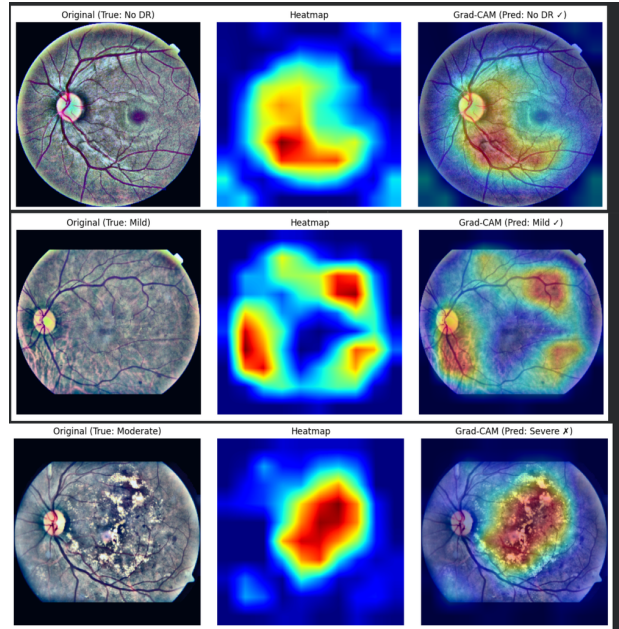


Figure 6: Grad-CAM visualizations highlighting the specific retinal regions driving model predictions.

7 Conclusion

Through systematic architectural and mathematical optimization, we developed a highly resilient DR grading pipeline. The transition from V1 to V2 augmentation resolved overfitting, while the integration of GeM ($p = 3$) pooling allowed for localized feature extraction. Finally, unfreezing 80% of the model parameters in conjunction with Conditional Ordinal Regression provided a mathematically sound, highly accurate grading system reflecting the true biological progression of Diabetic Retinopathy.